

LITERATURE REVIEW

DSN3099 – Engineering Projects in Community Service (EPICS) | Fall 2026–27
VIT Bhopal University

Project Title:	JanMitra Call – Government Scheme Eligibility and Guidance Helpline (IVR-based)
Broad Area:	ICT for Social Welfare – Digital Access to Government Services
Team Members:	Devraj Charan, Kajal Pareek, Tarun Yadav, Yash Burad, Suryansh Raghuwanshi, Utsang Kumar Jain

1. Context & Problem Formulation

Despite the rapid expansion of e-governance under the Digital India campaign and the establishment of comprehensive web repositories such as myScheme and the National Government Services Portal, a severe welfare access gap persists in rural India. Marginalized citizens, daily-wage laborers, and rural women frequently encounter prohibitive barriers including functional illiteracy, lack of smartphone access, unreliable cellular data connectivity, and steep digital UI learning curves. Knowing that a scheme exists is insufficient; beneficiaries encounter severe informational bottlenecks regarding documentary requirements, physical application venues, competent jurisdictional offices, and actionable next steps. This review systematically analyzes the technical and socio-technical literature across ICT for Development (ICT4D), telephony-based conversational interfaces, and applied Generative AI to substantiate the necessity and architecture of the JanMitra Call helpline.

2. Thematic Literature Survey

2.1 The Digital Divide, Intermediary Friction, and Information Asymmetry

Research by Pal et al. (2018) and Gupta et al. (2020) on digital public infrastructure in agrarian settings emphasizes that digitization without voice-first accessibility exacerbates social exclusion. Common Service Centres (CSCs) and Village Level Entrepreneurs (VLEs), while intended as physical access bridges, suffer from geographic clustering, informal fee gouging, and systemic delays. Citizens routinely travel significant distances only to be turned away due to missing documentation. The literature identifies that information asymmetry—specifically the absence of personalized eligibility evaluation and procedural clarity—is the primary obstacle preventing eligible households from obtaining Direct Benefit Transfers (DBT) and statutory welfare benefits.

2.2 Voice Interfaces & Historical IVR Systems in ICT4D

Spoken interaction is universally acknowledged as the most intuitive interface for non-literate and semi-literate demographics. Seminal field trials including Avaaj Otalo (Patel et al., 2010), CGNet Swara (Mudliar et al., 2012), and the IBM Spoken Web (Kumar et al., 2008) demonstrated that rural populations can readily utilize voice forums using basic 2G feature phones without formal training. However, these systems relied primarily on Dual-Tone Multi-Frequency (DTMF) touch-tone keypads ("Press 1 for agriculture, Press 2 for healthcare"). Human-Computer Interaction (HCI) research demonstrates that deep, branching DTMF hierarchies impose overwhelming cognitive load, resulting

in catastrophic call abandonment rates (>60%). Traditional DTMF cannot handle dynamic, multi-parametric eligibility inquiries where users must state age, land size, caste, family income, and residential block.

2.3 Recent Advances in Indian Language Speech Processing (ASR & TTS)

Historically, deploying natural conversational IVRs in India was constrained by severe ASR degradation caused by regional dialects, non-standard acoustic environments, and 8 kHz band-limited telephony channels (G.711 standard). Recent foundational work under Project Bhashini and AI4Bharat (Javed et al., 2022; Doddapaneni et al., 2023) has established state-of-the-art transformer-based acoustic and language models trained explicitly on 22 scheduled Indian languages, code-mixed data (e.g., Hinglish), and telephony-grade audio. These benchmarks confirm that modern sequence-to-sequence ASR and neural Text-to-Speech (TTS) pipelines have crossed the operational threshold necessary for intelligible, real-time public telephony interaction.

2.4 Conversational AI and Retrieval-Augmented Generation (RAG) for Public Policy

The integration of Large Language Models (LLMs) presents unprecedented potential for democratizing public knowledge. However, deploying off-the-shelf generative models in public governance introduces severe risks of confabulation (hallucination) and out-of-date policy dissemination. Lewis et al. (2020) and subsequent applied civic AI frameworks validate Retrieval-Augmented Generation (RAG) as the state-of-the-art solution. By strictly bounding the LLM’s generative context to a curated, vector-indexed database of verified government criteria, RAG completely mitigates ungrounded assertions while preserving fluent, empathetic conversational explanations in the citizen’s spoken dialect.

3. Comparative Evaluation of Existing Delivery Paradigms

The table below provides an analytical comparison of existing service delivery mechanisms against the proposed JanMitra Call system:

Service Model	Target Device	Literacy Requirement	Eligibility Flexibility	Operational Availability
Official Portals (myScheme)	Smartphone / PC	High (Read/Navigate)	Static form-based	24/7 (Requires Internet)
Kisan / Govt Helplines	Any Phone	Zero (Spoken)	Variable (Human Agent)	Restricted hours, High wait times
Civic WhatsApp Bots	Smartphone	Medium (Read/Text)	Rigid text prompts	24/7 (Requires Data)
JanMitra Call (Proposed)	Basic Feature Phone	Zero (Spoken Voice)	Dynamic LLM Reasoning	24/7 Scalable Automated Helpline

4. Research Gap and Engineering Justification

Despite extensive advancements in individual technological domains, a conspicuous integration gap exists between telecommunication accessibility and cognitive AI reasoning for social welfare: Existing high-intelligence tools remain locked behind smartphone screens and high literacy barriers, while universal voice channels remain constrained by obsolete DTMF keypad menus or expensive, unscalable call center workforces.

JanMitra Call addresses this exact systemic gap. By coupling carrier-grade IVR telephony (Twilio) with modern speech recognition and a Retrieval-Augmented Gemini AI backend, it provides a zero-barrier conversational gateway. Citizens simply state their circumstances naturally over a basic phone call, receive immediate eligibility determination with concrete administrative guidance, and receive an archival SMS summary for offline documentation.

5. Academic References

- [1] N. Patel, D. Chittamuru, A. Jain, P. Dave, and T. V. Parikh, "Avaaj Otalo: A field study of an interactive voice forum for small farmers in rural India," in Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI), 2010, pp. 733–742.
- [2] P. Mudliar, J. Donner, and W. Thies, "Citizen journalism meets telephony: The CGNet Swara experience," in Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI), 2012, pp. 1461–1470.
- [3] A. Kumar, N. Rajput, D. Chakraborty, S. Agarwal, and A. Nanavati, "The Spoken Web: An architectural perspective," ACM Transactions on Speech and Language Processing, vol. 8, no. 2, pp. 1–26, 2012.
- [4] J. Pal, A. Viswanathan, et al., "Agency in the Digital Age: Using Telephony and Voice for Civic Engagement in Developing Regions," Information Technologies & International Development, vol. 14, pp. 45–62, 2018.
- [5] T. Javed, S. Doddapaneni, A. Raman, et al., "Towards Building ASR Systems for the Next Billion Users," in Proceedings of the AAAI Conference on Artificial Intelligence, 2022, pp. 10813–10821.
- [6] S. Doddapaneni et al., "AI4Bharat-IndicSpeech: High quality open-source resources for Indic speech recognition and synthesis," in Proceedings of Interspeech, 2023, pp. 2450–2454.
- [7] P. Lewis, E. Perez, A. Piktus, et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in Advances in Neural Information Processing Systems (NeurIPS), vol. 33, 2020, pp. 9459–9474.
- [8] Ministry of Electronics and Information Technology (MeitY), Government of India, "Digital India Annual Report and National e-Governance Action Plan," Technical Report, New Delhi, 2023.
- [9] National Sample Survey Office (NSSO), "Household Social Consumption on Education and Digital Literacy in India," Ministry of Statistics and Programme Implementation, Report No. 585, 2022.